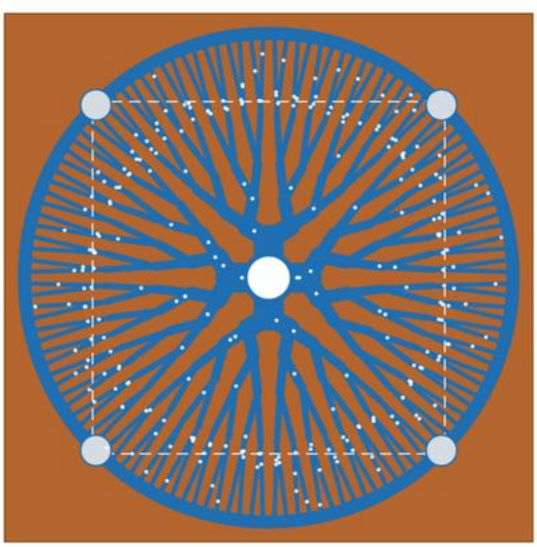
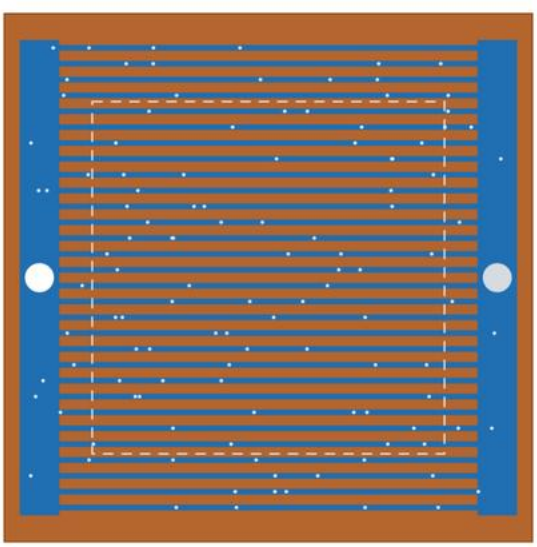
A leaf-shaped cold plate kept a 700 W chip 7 °C cooler.

Same copper, same water, same wetted area as 30 straight channels. Water and copper solved together in OpenFOAM.



Leaf-vein tree

peak case 66.1 °C



Straight channels

peak case 73.2 °C

57%
less pressure drop

15%
less entropy generated

8 / 8
checks passed, both plates

Bhuvan Nandipati

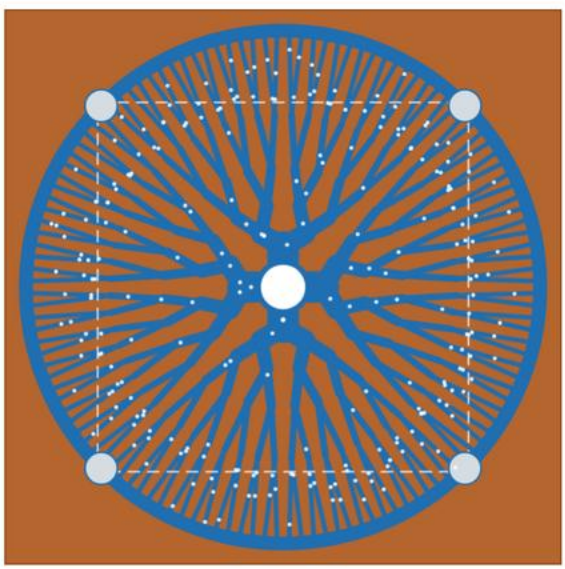
swipe →

Same copper, same water, same wetted area. The tree pumps easier and runs 7 K cooler.

A leaf-vein channel tree, laid out by a generalised Murray's law, against 30 straight channels. Both plates hold **2,084 mm³ of water** and wet **8,561 mm² of copper**. The question for the CFD was whether the cheaper flow path also runs cooler, or just pumps easier. The coarse-grid conjugate run says both: **2.3× less pressure drop** across the channel network and a **7.1 K lower peak case temperature** at 700 W. For an AI data center, the temperature result is the one that matters.

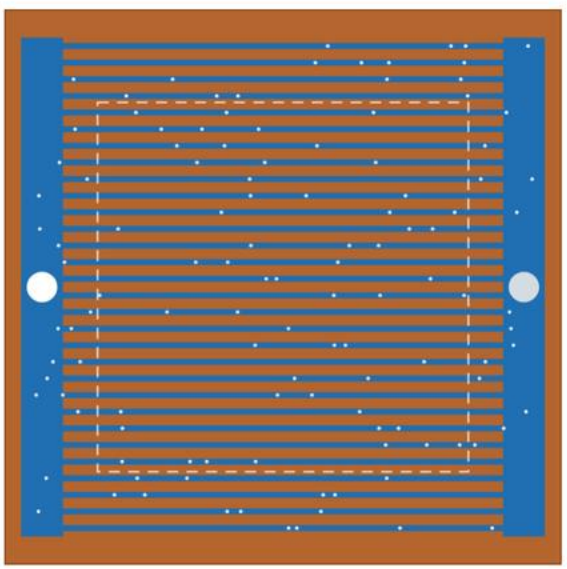
- Geometry built and checked
- 1-D screening model live below
- Conjugate CFD, coarse grid: 8/8 gates on both plates
- Grid study and turbulence check: Post 2

Leaf-vein tree — 99 Pa, 1-D



4 → 128 channels over 6 generations, 3.42 mm down to 0.50 mm wide, 2.0 mm deep. Inlet at the hub, four outlets on the ring.

Parallel channels — 514 Pa, 1-D



30 channels, 0.643 mm wide, 2.0 mm deep, fed from end manifolds. Volume and wetted area matched to the tree within 0.03%.

copper C110, 60 × 60 × 5 mm · coolant (dots move at the real relative speeds, slowed about 25×) · 40 × 40 mm die

terminal channel 0.13 m/s · parallel channel 0.43 m/s

05 What the CFD answered

Claims stated before the run, answered by the coarse-grid run of 9 October 2026. Conjugate CHT in OpenFOAM v2512 (chtMultiRegionSimpleFoam), laminar, steady, copper and water solved together. The tree is a quarter model (1.11 M cells), the parallel plate a half model (2.42 M cells); both carry 700 W per full plate.

Δp, plenum to plenum

2.34 × lower

696 Pa vs 1,630 Pa

Peak case temperature

−7.1 K

66.1 °C vs 73.2 °C

Entropy generated, whole plate

1.18 × lower

0.106 vs 0.126 W/K

Energy balance

0.07 % worst

175.0 / 175 W · 349.8 / 350 W

Quantity, 700 W, 1.0 L/min, 40 °C inlet	Tree	Parallel	Ratio
Δp, plenum to plenum	696.2 Pa	1,630.2 Pa	2.34×
Δp, port to port (lower bound: laminar ports)	1,100.6 Pa	5,014.6 Pa	4.56×
Peak case temperature	66.10 °C	73.21 °C	−7.1 K
Mean case temperature	63.89 °C	66.06 °C	−2.2 K
Coolant temperature rise	10.15 K	10.14 K	same
Entropy generation, whole plate	0.1064 W/K	0.1255 W/K	1.18×
Viscous entropy generation, whole plate	1.22×10^{-5} W/K	3.08×10^{-5} W/K	2.52×
Bejan number	0.99989	0.99976	
Closure, volume vs. boundary entropy	−13.5 %	−7.9 %	
Steadiness, entropy band over the last 100 samples	0.68 %	0.78 %	

Claim	Screening (1-D)	CFD (coarse)	What it means
Tree cuts channel Δp	5.18×	**2.34×**	Holds, at about half the screening ratio. Junction and plenum losses, which the 1-D model omits, take the rest. Port to port it is 4.56×
Case temperature at 700 W	< 72.4 °C	**66.1 / 73.2 °C peak**	Means are below the ceiling on both plates. The 7.1 K gap in the peaks is the heat-spreading effect the 1-D model cannot see
Loss is almost all thermal (Bejan ≈ 1)	0.99997	**0.99989**	Holds. Friction is about 0.01% of the loss; the plate's loss is heat crossing a temperature difference
Wall shear even across generations (Murray invariant)	24.6% spread	**164% spread**	Breaks. Developing flow puts high shear at every branch entry; the developing-flow estimate was 185%
Volume entropy = boundary ledger (closure)	0.174 W/K	**−13.5% / −7.9%**	Within the 15% coarse-grid limit. The tree sits close to it, so its entropy numbers carry that uncertainty until the grid study

Eight gates every run must pass before any number counts: all passed on both plates

G1

Mesh quality

non-ortho < 65°, no negative cells

- pass · tree / parallel: 64.96° / 45.41°

G2

Interface match

fluid and solid faces agree exactly

- pass · tree / parallel: 198,212 / 355,474 faces

G3

Plenum zones

both zones contain cells

- pass · tree / parallel: 1,368 + 2,472 / 1,572 + 1,577

G4

Mass balance

inlet = outlet within 0.1%

- pass · tree / parallel: 0.0000% / 0.0002%

G5

Flow rate

matches the spec within 1%

- pass · tree / parallel: 0.000% / 0.000%

G6

Energy balance

$\dot{m} \cdot cp \cdot \Delta T$ = heater power within 1%

- pass · tree / parallel: 0.008% / 0.073%

G7

Positive entropy

no cell with $S''' < 0$

- pass · tree / parallel: min 0 / min 0

G8

Closure

volume vs. boundary within 15% on the coarse grid, 10% from Post 2

- pass · tree / parallel: −13.5% / −7.9%

What this run doesn't prove yet

- It is one coarse mesh. The 7 °C gap has to survive a finer one.
- The tree's entropy check closes at −13.5%, inside my 15% coarse-grid limit but close to it.
- Flow at the ports is near-turbulent, so port-to-port pressure drop is a lower bound.

Post 2: a grid study and a turbulence check. Does the 7 °C hold?

Try it yourself: the interactive version, with both plates, sliders and all eight checks, is linked in the comments.

Your turn: if you design cold plates, could a branching layout like this be made at scale? Or is manufacturing where it falls apart?